Entity–Relationship Diagram

Bank Account and Transaction Management System · Project 55 (DBMS PBL)

MySQL 8 · InnoDB · 3NF

15 tables · 19 foreign keys · crow's-foot (IE) notation · source: sql/01_schema.sql

BRANCH

PK	**branch_id**	INT
UK	ifsc_code	CHAR(11)
	branch_name	VARCHAR(100)
	address	VARCHAR(255)
	city	VARCHAR(60)
	phone	VARCHAR(15)
	created_at	DATETIME

ACCOUNT_TYPE

PK	**account_type_id**	INT
UK	type_code	VARCHAR(20)
	type_name	VARCHAR(60)
	category	ENUM
	normal_side	ENUM
	min_opening_balance	DECIMAL(15,2)
	min_balance	DECIMAL(15,2)
	max_holders	TINYINT
	interest_rate	DECIMAL(5,2)
	description	VARCHAR(255) NULL

CUSTOMER

PK	**customer_id**	INT
	first_name	VARCHAR(50)
	last_name	VARCHAR(50)
	date_of_birth	DATE
	gender	ENUM
UK	email	VARCHAR(100)
	phone	VARCHAR(15)
	address	VARCHAR(255)
	city	VARCHAR(60)
UK	id_proof_type	ENUM
UK	id_proof_number	VARCHAR(20)
	created_at	DATETIME
	updated_at	DATETIME

LOAN

PK	**loan_id**	INT
UK	loan_number	CHAR(14)
FK	account_id	INT
FK	customer_id	INT
	loan_type	ENUM
	principal	DECIMAL(15,2)
	annual_interest_rate	DECIMAL(5,2)
	tenure_months	SMALLINT
FK UK	disbursement_txn_id	BIGINT
FK	disbursement_txn_type	ENUM

ACCOUNT_STATUS_HISTORY — append-only

PK	**history_id**	BIGINT
FK	account_id	INT
	old_status	ENUM NULL
	new_status	ENUM
	reason	VARCHAR(255)
	changed_at	DATETIME

ACCOUNT

PK	**account_id**	INT
UK	account_number	CHAR(12)
FK	branch_id	INT
FK	account_type_id	INT
	balance	DECIMAL(15,2)
	status	ENUM
	opened_at	DATETIME
	closed_at	DATETIME NULL

ACCOUNT_HOLDER — associative (M:N)

PK FK	**account_id**	INT
PK FK	**customer_id**	INT
	holder_role	ENUM
	added_at	DATETIME
UK	*primary_account_id*	INT NULL GEN

BENEFICIARY

PK	**beneficiary_id**	INT
FK UK	customer_id	INT
FK UK	beneficiary_account_id	INT
UK	nickname	VARCHAR(50)
	status	ENUM
	added_at	DATETIME
	deactivated_at	DATETIME NULL

LOAN_SCHEDULE — weak entity

PK FK	**loan_id**	INT
PK	**installment_no**	SMALLINT
	due_date	DATE
	principal_component	DECIMAL(15,2)
	interest_component	DECIMAL(15,2)
	installment_amount	DECIMAL(15,2) GEN

TRANSACTION_ENTRY — ledger line

PK	**entry_id**	BIGINT
FK UK	transaction_id	BIGINT
FK UK	account_id	INT
UK	entry_side	ENUM
	amount	DECIMAL(15,2)
	balance_after	DECIMAL(15,2)

BANK_TRANSACTION — ledger header (supertype)

PK	**transaction_id**	BIGINT
UK	reference_no	CHAR(16)
UK	transaction_type	ENUM
	status	ENUM
	narration	VARCHAR(255) NULL
FK UK	reversal_of_id	BIGINT NULL
	posted_at	DATETIME

TRANSFER — subtype

PK FK	**transaction_id**	BIGINT
FK	transaction_type	ENUM
FK	beneficiary_id	INT NULL
	remarks	VARCHAR(140) NULL

CHARGE — subtype

PK FK	**transaction_id**	BIGINT
FK	transaction_type	ENUM
FK	charge_type_id	INT

CHARGE_TYPE

PK	**charge_type_id**	INT
UK	charge_code	VARCHAR(30)
	charge_name	VARCHAR(80)
	default_amount	DECIMAL(15,2)

LOAN_REPAYMENT — subtype

PK FK	**transaction_id**	BIGINT
FK	transaction_type	ENUM
FK UK	loan_id	INT
FK UK	installment_no	SMALLINT

hosts · classifies · holds · saves · borrows · has · is held by · is payee of · disburses · posted to · authorises · consists of · is a · reverses · categorises · settled by · is a

LEGEND

Organisation & products · Customers & accounts · Double-entry ledger · Loans

Subtypes TRANSFER, CHARGE and LOAN_REPAYMENT are disjoint specialisations of BANK_TRANSACTION (composite FK on transaction_id + transaction_type). Every FK is ON DELETE RESTRICT.

exactly one · zero or one · one or more · zero or more

PK primary key (underlined) · FK foreign key · UK unique (alone or composite) · NULL nullable · GEN generated column (italic)

identifying relationship · non-identifying · double border = weak entity